From Raw Shots to Expectation Values: Shot Grouping Improves Time Series Prediction

Markus Baumann*, Maximilian Zorn*, Thomas Gabor†, Claudia Linnhoff-Popien*, and Jonas Stein*
*QAR-Lab, Department of Computer Science, LMU Munich, Munich, Germany
†Department of Computer Science, University of Exeter, Exeter, United Kingdom

***Abstract*—Between keeping every shot as a noisy training example and averaging all shots into one clean feature lies an entire spectrum of data organizations that quantum machine learning usually leaves implicit. We make that choice explicit. Shot grouping partitions a fixed measurement record into disjoint groups and averages within each group, tuning smoothly between the two conventions with a single validated parameter and no additional quantum executions. We evaluate the method on chaotic synthetic benchmarks and on real-world data under strict execution budgets. An intermediate grouping consistently outperforms the standard expectation-value pipeline. Matched controls show that the gain acts as correctly scaled regularization of the classical readout. Unmitigated runs on two superconducting processors confirm the trend on every task. Shot grouping therefore offers a practical way to improve finite-shot quantum learning at no added quantum cost.**

***Index Terms*—Quantum computing, Machine learning, Reservoir computing, Time series analysis**

I. Introduction

Every quantum learning experiment ends the same way: a circuit runs many times, and each run returns one measured bitstring. Before anything can be learned, this pile of shots must become training data. The step looks like bookkeeping, and it is usually treated that way. Almost every pipeline averages all shots of a time step into a single expectation-value estimate and moves on. This paper asks what that habit costs.

The habit has a respectable origin. Expectation values are the native currency of quantum mechanics: they are what theory predicts and what experiments are built to estimate. A learning pipeline, however, does not consume observables. It consumes training examples, and it cares about two things only: how many there are and how noisy each one is. The moment a measurement record becomes training data, a physics convention quietly turns into a statistical decision: spend all measurements on making one example clean, or make several examples that are merely clean enough. No single choice can suit every task, budget, and noise level.

Averaging everything is one end of a dial. It buys the cleanest possible feature vector but pays with scarcity. Thousands of circuit executions collapse into a handful of training examples, and too few examples relative to the number of features destabilizes regression [1]. The opposite end, keeping every shot as its own example, buys abundance but pays with noise, and noisy features systematically shrink fitted coefficients [2]. Between these extremes lies an entire range of aggregations.

Corresponding author: markus.baumann@campus.lmu.de.

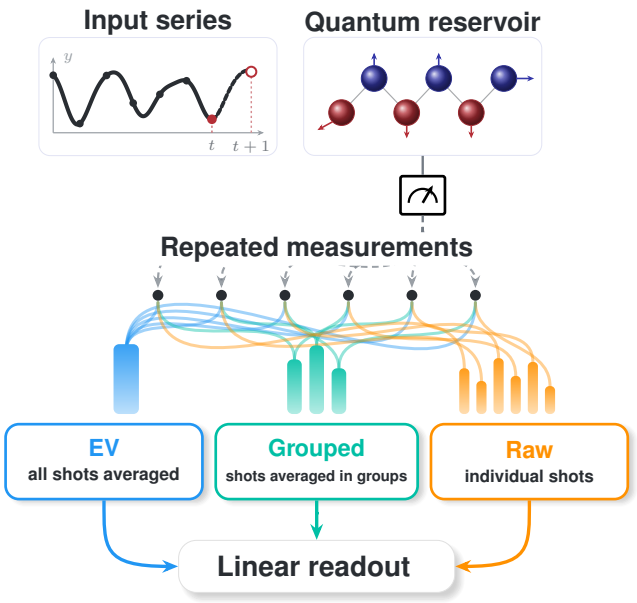


Fig. 1. Three aggregation modes for one finite measurement record. A normalized input drives a fixed quantum reservoir. The same repeated measurements are processed using full expectation-value averaging (EV), disjoint group averaging (Grouped), or individual shots (Raw). Only the classical aggregation of the measurement record changes.

Nothing in the record forces either convention, and the record does not care which one tradition prefers.

Quantum reservoir computing (QRC) is a clean place to study this dial [3]–[5]. A fixed quantum circuit transforms an input sequence into measured features, and only a classical readout is trained. The circuit, the targets, and the measurement record can therefore be held completely fixed while the aggregation alone changes. Whatever happens to the forecast is attributable to the aggregation because nothing else moved.

We compare three representations of one record (Fig. 1). The expectation-value representation (EV) averages all measurements at each time step. Raw uses each measurement separately. Grouped averages disjoint subsets, and its aggregation scale k, the number of shots per averaged training view, slides smoothly from $k = 1$ (Raw) to $k = N$ (EV).

The dial turns out to matter. Under one matched protocol and fixed execution budgets, an intermediate aggregation lowers test error relative to full averaging on every task we evaluate, including two synthetic systems and two real-world datasets.

The best aggregation scale depends on the record, so no single convention is appropriate. The gain requires nothing quantum. Matched controls trace it to correctly scaled regularization of the classical readout, supplied automatically by the record's own fluctuations. Because aggregation is classical post-processing, it costs no additional quantum executions and applies retroactively to any archived record.

Our contribution is *adaptive shot aggregation*. The number of shots per training view becomes a setting of the classical readout, selected by validation like any other hyperparameter, at no additional quantum cost. We map when the choice matters through three budget allocations of time steps and shots. We explain why it works through matched controls that isolate the regularization mechanism. We test where it transfers through unmitigated experiments with a separate hardware reservoir on two superconducting processors.

II. Related Work

Finite sampling can be addressed at three points in a quantum learning pipeline. It can be addressed before execution, by deciding how to spend a measurement budget. It can be addressed after averaging, by correcting the statistics of expectation values. Or it can be addressed at the record itself, by changing how repeated measurements become training data. Our method lives at this third point, which is the least studied of the three.

The idea of averaging subsets of measurements is not new, and we do not claim it. Cemin *et al.* average subsets of measurements to enlarge a training set for reduced quantum channel learning [6], and it is a classical result that training with noisy inputs behaves like an additional penalty on large fitted weights [7]. What has been missing is the treatment of the aggregation scale as a tunable quantity. This study evaluates the complete range from Raw to EV on one fixed QRC record, calibrates the scale jointly with the readout regularization, gives an exact account of the penalty this creates, and tests the result under three allocations of a fixed execution budget on four matched forecasting tasks. The contribution is the calibration and evaluation of the aggregation scale, not the original introduction of subset averaging.

The other two points of the pipeline are well developed and complementary to ours. Before execution, resource-aware optimization decides how many measurements each part of a computation deserves [8]. After averaging, Ahmed *et al.* derive noise-aware training rules for expectation values [9], and Kreplin and Roth reduce sampling noise through variance regularization [10]. Fundamental limits of learning from finitely sampled physical systems have also been mapped out [11]. None of these methods competes with ours. Allocation ends where our method begins, and the averaged quantities that noise-aware corrections operate on can themselves be produced by grouped aggregation. Our method needs nothing beyond a single archived measurement record.

Within QRC, a rich line of work studies how noise, dissipation, measurement, finite Hilbert space, and input transformations shape temporal features [12]–[23]. All of these studies analyze or modify the quantum representation. We do the opposite. The representation is held completely fixed, and only the organization of its repeated measurements changes. Because every reservoir in those studies still produces a measurement record, adaptive shot aggregation applies on top of any of them.

III. Method

A. Simulation reservoir

The reservoir is a fixed six-qubit circuit that turns each scalar input into features. It sees the input twice, through two shallow reuploading layers, which let a shallow circuit represent richer functions of the input [24]. In each layer every qubit is first rotated by an input-dependent amount, using R_y and R_z with a fixed per-qubit offset. Neighboring qubits are then entangled by R_{zz} couplings and mixed by a small fixed non-Clifford rotation, keeping the circuit outside the Clifford-only setting. All angles are drawn once and frozen, so the circuit never changes during forecasting, and several independent frozen circuits check that the results do not depend on one lucky draw. A single computational-basis measurement then yields eleven features at once: the six qubit signals Z_i and the five neighbor correlations Z_iZ_{i+1}. A leaky integrator with leak 0.2 and the ten most recent steps add temporal memory, and a ridge regression maps the features to the next value, scored by test NRMSE.

B. Adaptive shot aggregation

Write $x_t^{(n)}$ for the features produced by shot n at time t. We split the N shots of each step into $G = N/k$ groups of equal size k and average within each group,

$$x_t^{(g)} = \frac{1}{k}\sum_{n\in g} x_t^{(n)}, \qquad g = 1, \ldots, G. \tag{1}$$

Every group is a partly averaged view of the same time step, and all views share the same target y_t. At test time we average the G predictions of a step, which is identical to predicting once from the averaged features. The two methods are therefore scored in exactly the same way, so any difference between EV and Grouped is created during training, not at test time.

The shot index carries no physical meaning, because shot n at one step and shot n at the next come from unrelated circuit runs. We therefore shuffle the shots freshly at every time step before forming groups, and we repeat the whole analysis over several random groupings.

C. Fixed execution budget and training

Evaluating T time steps with N shots per step costs TN circuit executions. We hold this budget fixed at ten thousand executions and study three ways of spending it, $(T, N) = (250, 40)$, $(500, 20)$, and $(1000, 10)$, ranging from fewer time steps with more shots to more time steps with fewer shots. For each setting we try every group size k that divides N, which includes Raw at $k = 1$ and EV at $k = N$.

Using three expanding splits, each candidate pair of group size and ridge strength (k, λ) is fitted on an earlier block and

scored on the block that follows. The best pair is then frozen, the readout is refitted on all data before the test block, and the final fifth of the series is scored once. Normalization and feature scaling use training data only, so the test block never influences any choice.

D. Why grouping regularizes the readout

Full averaging hands the readout one clean picture of each time step. Individual shots hand it many pictures, each blurry. Intermediate groups hand it a few lightly cleaned pictures of the same target, and the differences between those pictures reveal which parts of the feature vector are stable and which are only measurement noise. Fitting the readout to all of the views at once discourages it from trusting the unstable parts.

For a linear readout with squared loss this becomes an exact identity. Let $\bar{x}_t$ be the average of the G grouped views, and let $C_{k,t} = \frac{1}{G}\sum_g (x_t^{(g)} - \bar{x}_t)(x_t^{(g)} - \bar{x}_t)^\top$ record how those views scatter around their average, that is, how the features wobble together from group to group. For any intercept b and weights w,

$$\frac{1}{G}\sum_g \left(y_t - b - w^\top x_t^{(g)}\right)^2 = \left(y_t - b - w^\top \bar{x}_t\right)^2 + w^\top C_{k,t}\, w. \quad (2)$$

The first term on the right is exactly the EV loss. The second charges extra whenever the weights lean on feature combinations that vary strongly across the views. Grouped training is therefore EV training plus a penalty read directly off the observed shot fluctuations, and the identity assumes neither Gaussian noise nor independence over time.

The identity motivates a matched baseline: covariance-regularized expectation-value training (Cov-EV). Cov-EV keeps the fully averaged EV features and adds a penalty estimated from how the individual shots fluctuate together. Its strength and the ordinary ridge strength are chosen using the same earlier data as the other methods. Cov-EV does not reuse the aggregation scale chosen by Grouped; it tests whether the same regularization mechanism can be applied directly after full averaging. Similar performance from Grouped and Cov-EV would therefore support regularization as the source of the gain.

The equivalence is also verified computationally. When k is fixed, Grouped training and EV training with the penalty defined by $C_{k,t}$ give identical predictions. Because the leaky integrator and overlapping lags make neighboring feature vectors dependent, we report uncertainty across trajectories or measured windows rather than across the many correlated grouped views.

IV. Experimental Setup

A. Simulation setup

We evaluate Mackey–Glass [25], Rössler [26], Santa Fe competition Series A far-infrared laser intensity [27], and monthly sunspots [28]. In simulation, all four tasks use the same frozen six-qubit reservoir, $Z + ZZ$ features, leak, ten lags, washout, training based scaling, time-ordered model selection, execution budgets, and candidate settings. The only task specific inputs are the source sequence and forecast window origins.

Each synthetic task contributes five independently generated trajectories. Each real-world dataset contributes five chronological windows with disjoint test blocks. Three circuit parameter settings, three measurement samples, and three grouping assignments repeat the experiment within each forecasting unit. We first average these repetitions within each unit and then compute paired 95% intervals across the five separate trajectories or windows. This prevents repeated simulations of one sequence from being counted as independent forecasting problems.

The main comparison includes the identically leaked and lagged scalar input, ideal reservoir features without sampling noise, EV, Grouped, Raw, and independently tuned Cov-EV training. Every sampled method reuses the identical raw record, time-ordered splits, ridge candidates, and test access policy. Twenty extra grouping assignments at the balanced setting test sensitivity to arbitrary group assignments.

After a washout of 30 steps, the last 20% of each usable sequence is reserved for testing and is not inspected while settings are chosen. The earlier block is evaluated in three rounds: each round fits on the past and checks the forecast on the next block. For every aggregation scale k, the ridge value is chosen from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$; Cov-EV uses the same ridge values with eight predeclared strengths for the fluctuation-based penalty. Scaling is recomputed using only the fitting portion in every round. Scoring first averages all Grouped predictions back to one forecast per time step. Thus every method contributes the same number of forecasts to the comparison. The test block is accessed only after all choices have been fixed.

The allocation $(T, N) = (500, 20)$ is the primary comparison. The other two allocations use the same budget of 10,000 executions and are secondary analyses. Reservoir parameters, generators, window origins, repeat identifiers, grouping rules, candidate settings, and reporting filters are stored before the test blocks are evaluated. The reservoir, aggregation choices, ridge choices, and test procedure are identical across tasks.

B. Hardware setup

The device study intentionally uses a different reservoir design. Its fixed ten-qubit circuit contains one amplitude-encoded input qubit, nine Hadamard-initialized memory qubits, one R_{ZZ} ring layer, transverse R_X rotations, and ten local-Z outputs. The circuit is mapped to selected physical qubits on *ibm_fez* and *ibm_marrakesh* and is held fixed across all four hardware tasks. It tests whether the same aggregation procedure transfers to a second circuit and set of measured features; it is not a hardware implementation of the six-qubit simulator or a qubit-count comparison. Within each study, every aggregation method receives the same circuit outputs and only the classical organization of shots changes. One $T = 500$, 400-shot acquisition per task and processor is partitioned into twenty ordered and disjoint records with $N = 20$. Hardware intervals describe variation among measurement blocks within

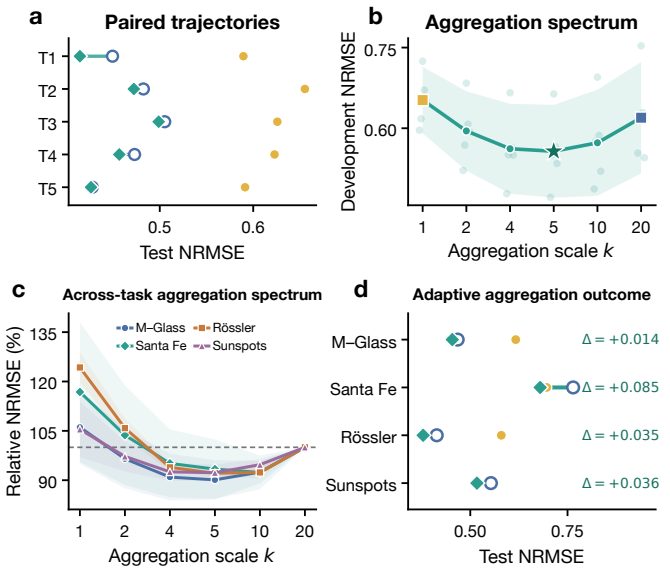


Fig. 2. Summary under the shared protocol. (a) Test outcomes for Mackey–Glass for five independent trajectories. (b) Balanced aggregation spectrum over aggregation scale. (c) The spectrum relative to EV across tasks. (d) Test outcomes after choosing aggregation scale and ridge strength together. Panels (b,c) use development data only; panels (a,d) use test blocks. Bands and intervals use separate trajectories or windows; faint marks show repeated measurement trials within them.

one acquisition. They do not represent independent devices or calibration periods.

V. Evaluation

We report accuracy as test NRMSE, a normalized error for which lower is better and where one means predicting the mean. Three ways of turning the shot record into features are compared throughout. EV averages all shots of a step into one clean feature vector. Raw keeps every shot as its own noisy example. Grouped is our method and sits between them. A fourth entry, Cov-EV, is a control that adds an independently tuned covariance penalty to plain averaging. We summarize the benefit of Grouped as the paired difference EV minus Grouped, so a positive value means Grouped is more accurate. Each difference comes with a 95% confidence interval. An interval above zero provides evidence of a positive mean improvement; an interval containing zero is inconclusive. The balanced setting spends the fixed budget on 500 time steps with 20 shots each.

A. Balanced setting

At the balanced budget, Grouped lowers mean NRMSE relative to EV on all four tasks (Table I, Fig. 2). The improvement is 0.0138 on Mackey–Glass (95% CI $[-0.0023, 0.0300]$), 0.0848 on Santa Fe laser ($[0.0174, 0.1522]$), 0.0349 on Rössler ($[0.0139, 0.0559]$), and 0.0360 on sunspots ($[0.0103, 0.0616]$). The interval stays above zero on the two real-world datasets, Santa Fe laser and sunspots, and on Rössler. Only Mackey–Glass is inconclusive.

Santa Fe laser shows the largest effect. Grouped reduces the error from 0.764 to 0.680, and all five disjoint test windows favor Grouped. When k and the ridge strength are tuned

TABLE I

Mean test NRMSE at the balanced setting. Repeated measurement trials are averaged within each of five separate trajectories or windows. Lower values are better.

Task	Class.	Ideal	EV	Grouped	Raw	Cov-EV
Mackey–Glass	0.100	0.178	0.468	0.454	0.617	**0.447**
Santa Fe laser	0.762	0.493	0.764	0.680	0.697	**0.677**
Rössler	0.096	0.054	0.414	0.379	0.580	**0.367**
Sunspots	0.387	0.474	0.553	0.517	0.558	**0.515**

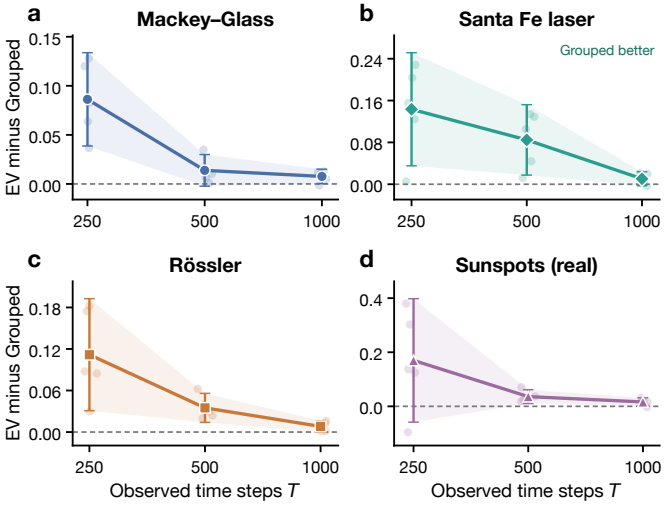


Fig. 3. Comparison at fixed execution budget. Each panel shows $\Delta = \mathrm{NRMSE}_{\mathrm{EV}} - \mathrm{NRMSE}_{\mathrm{Grouped}}$ at $(T, N) = (250, 40), (500, 20), (1000, 10)$. Positive values favor Grouped. Points are trajectory or window units; markers and bars are means and 95% intervals.

together, the search lands on $k = 10$ most often and on an intermediate scale, neither Raw nor EV, in 94% of repeated trials. Cov-EV reaches 0.677, almost the same as Grouped, which already suggests that the gain is a regularization effect rather than new physical information. The gain also depends on how the budget is spent. On Santa Fe it shrinks from 0.143 with 40 shots per step to 0.010 with 10 shots per step because grouping has more shots to reorganize when shots are plentiful.

We also compare an ideal noise-free reservoir with a classical baseline that uses identical preprocessing. The quantum reservoir is worse on Mackey–Glass and sunspots, better on Santa Fe, and inconclusive on Rössler. Our claims are therefore about organizing a chosen reservoir record, not quantum advantage. The preferred aggregation scale varies across tasks, trajectories, circuit settings, measurement draws, and grouping assignments, and EV is occasionally best. Equation (2) explains this variation: changing k changes which feature combinations the readout is discouraged from using, not merely the strength of that regularization. Raw is less accurate than Grouped on all four task means.

B. Trading time steps against shots

The fixed budget of ten thousand executions can be spent in different ways, and Fig. 3 compares three of them. When each of 250 time steps receives 40 shots, Grouped improves every task, with mean gains from 0.086 to 0.170. The interval stays

above zero for Mackey–Glass, Santa Fe laser, and Rössler, while the sunspot result varies across windows. When the same budget is spread over 1000 time steps with only 10 shots each, the gains shrink to between 0.008 and 0.016, and the interval stays above zero only for Mackey–Glass and sunspots. Grouping helps most when a step is measured many times and the series is short, because full averaging then produces very few training views from a shot-rich record.

C. Why the gain is regularization

The central control is Cov-EV, which keeps EV features and adds a penalty based on how the individual shots fluctuate together. At the balanced setting Cov-EV has a slightly lower mean error than Grouped on all four tasks, by 0.0021 to 0.0116 NRMSE, and only the Rössler interval excludes zero. Grouped obtains its penalty directly when the views are formed, whereas Cov-EV separately chooses the covariance-penalty strength on earlier data. Their near-match is the main evidence that the benefit described by Eq. (2) comes from regularizing the readout rather than from extra information from the quantum device.

Two further checks support this interpretation. Raw is less accurate than EV on Mackey–Glass and Rössler and inconclusive on the two real-world datasets, so noisy single shots are not simply better data. The exact two-term form of Eq. (2) also reproduces the grouped ridge fit at a fixed k to numerical precision. Cov-EV is a separate control: it uses the full shot covariance and independently chooses its strength.

Grouping is also robust to how the groups are drawn. Across twenty repeated settings the average improvements are -0.0064 on Mackey–Glass, 0.1103 on Santa Fe laser, 0.0177 on Rössler, and 0.0493 on sunspots. Only the sunspot interval stays above zero, reflecting the wide variation across settings. Fixing one grouping across time instead of reshuffling at every step changes the error by at most 0.0021 NRMSE, with every interval containing zero. The grouping convention therefore matters far less than ordinary run-to-run variation. Figure 4 collects these controls.

D. Transfer to hardware

The same experiment was run without error mitigation on two superconducting processors, *ibm_fez* and *ibm_marrakesh* (Fig. 5). Grouped improves every task on both machines, with mean gains from 0.038 to 0.153 NRMSE on *ibm_fez* and from 0.031 to 0.103 on *ibm_marrakesh*. On Santa Fe laser, the error falls from 0.750 to 0.711 on *ibm_fez*, a gain of 0.038 ($[0.027, 0.050]$), and from 0.744 to 0.713 on *ibm_marrakesh*, a gain of 0.031 ($[0.023, 0.039]$). Grouped is more accurate in 19 of 20 measurement blocks on each processor.

These intervals describe variation among the twenty measurement blocks of a single acquisition. They do not measure variation across independent hardware campaigns, so the hardware results demonstrate consistent transfer of the effect rather than a precise device-level comparison.

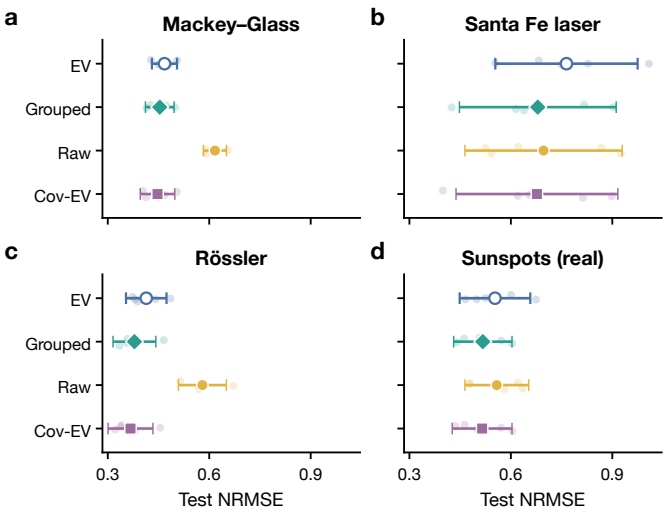


Fig. 4. Controls at the balanced setting in the order EV, Grouped, Raw, and Cov-EV. Faint points are the five trajectory or window means after repeated measurement trials are averaged within each unit; markers and bars show means and 95% intervals.

VI. Discussion

Adaptive shot aggregation changes only the classical use of a measurement record. The aggregation scale k determines how many shots form each training view, while Raw and EV remain available as its two endpoints. Earlier time blocks choose k and the ridge strength before the test block is evaluated.

The method requires no additional circuit executions and is

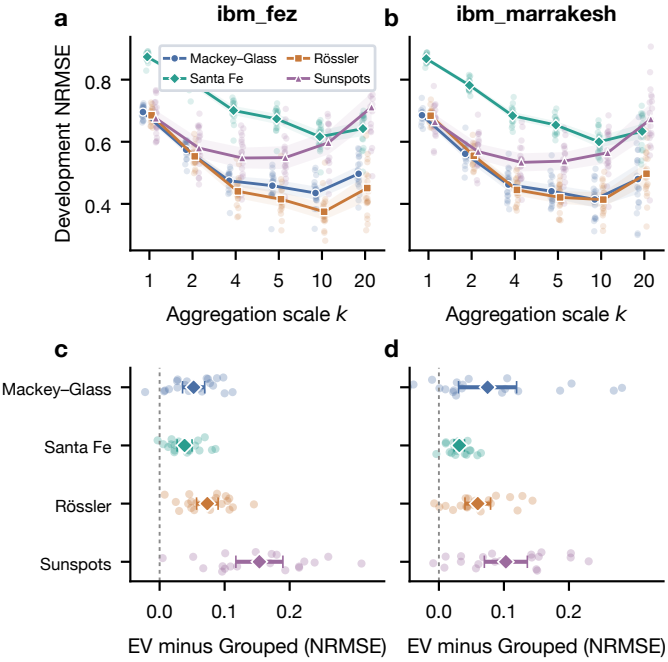


Fig. 5. Unmitigated transfer results from the separate ten-qubit reservoir on *ibm_fez* and *ibm_marrakesh*. For each task and processor, one record with $T = 500$ and 400 shots per step is partitioned into twenty disjoint records with $N = 20$. The upper panels show performance across aggregation scales. The lower panels show the test difference between EV and Grouped; positive values favor Grouped.

most useful when a record contains many shots per time step but relatively few time steps. In this regime, partial averaging supplies several denoised views of each target. Equation (2) shows that these views impose a penalty derived from the measured shot fluctuations. Cov-EV obtains slightly lower mean errors but remains close to Grouped, supporting covariance regularization as the mechanism while showing that grouping is not its only implementation.

The positive trend transfers from the six-qubit simulation reservoir to a separate ten-qubit circuit on two processors. This comparison tests transfer of the aggregation procedure, not the effect of qubit count. Alternative scales require individual measurement outcomes and cannot be recovered from stored expectation values. The hardware intervals describe blocks from one acquisition per task and processor; independent campaigns and other reservoir families are the next tests.

VII. Conclusion

Turning a shot record into training data has always involved a choice, and the field has answered it by reflex, averaging everything. We show the reflex is beatable. Choosing the aggregation scale by validation lowers mean test error on chaotic and real-world forecasting at the balanced budget, holds on two superconducting processors, and requires no additional quantum cost. Because every measurement-based reservoir produces such a record, the choice we make explicit here is available to any of them, and the question it raises belongs to finite-shot learning in general.

Appendix A
Reproducibility Summary

All simulation tasks use the same six-qubit reservoir, preprocessing, chronological selection procedure, and execution budgets. Mackey–Glass and Rössler use five trajectories; Santa Fe and sunspots use five windows with disjoint test blocks. Repeated settings are averaged within each unit before uncertainty is computed. All methods share the raw record and ridge candidates. The three budgets are $(T, N) = (250, 40), (500, 20), (1000, 10)$, each totaling 10,000 executions. Hardware uses one fixed ten-qubit circuit and one 400-shot record per task and processor, split into twenty disjoint $N = 20$ blocks. The supplement and artifactprovide all splits, settings, outputs, hardware job IDs, checksums, and rebuild scripts.

References

[1] D. Hsu, S. M. Kakade, and T. Zhang, "Random Design Analysis of Ridge Regression," in *Proceedings of the 25th Conference on Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 23, 2012, pp. 9.1–9.24.

[2] W. A. Fuller, *Measurement Error Models*. New York: Wiley, 1987.

[3] K. Fujii and K. Nakajima, "Harnessing Disordered-Ensemble Quantum Dynamics for Machine Learning," *Physical Review Applied*, vol. 8, no. 2, p. 024030, 2017.

[4] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa, "Boosting Computational Power through Spatial Multiplexing in Quantum Reservoir Computing," *Physical Review Applied*, vol. 11, no. 3, p. 034021, 2019.

[5] P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi *et al.*, "Opportunities in Quantum Reservoir Computing and Extreme Learning Machines," *Advanced Quantum Technologies*, vol. 4, no. 8, p. 2100027, 2021.

[6] G. Cemin, M. Cech, E. Weiss, S. Soltan, D. Braun *et al.*, "Machine Learning of Reduced Quantum Channels on Noisy Intermediate-Scale Quantum Devices," *Physical Review A*, vol. 110, no. 5, p. 052418, 2024.

[7] C. M. Bishop, "Training with noise is equivalent to Tikhonov regularization," *Neural Computation*, vol. 7, no. 1, pp. 108–116, 1995.

[8] C. Moussa, M. H. Gordon, M. Baczyk, M. Cerezo, L. Cincio, and P. J. Coles, "Resource Frugal Optimizer for Quantum Machine Learning," *Quantum Science and Technology*, vol. 8, no. 4, p. 045019, 2023.

[9] O. Ahmed, F. Tennie, and L. Magri, "Optimal Training of Finitely Sampled Quantum Reservoir Computers for Forecasting of Chaotic Dynamics," *Quantum Machine Intelligence*, vol. 7, p. 31, 2025.

[10] D. A. Kreplin and M. Roth, "Reduction of Finite Sampling Noise in Quantum Neural Networks," *Quantum*, vol. 8, p. 1385, 2024.

[11] F. Hu, G. Angelatos, S. A. Khan, M. Vives, E. Türeci *et al.*, "Tackling Sampling Noise in Physical Systems for Machine Learning Applications: Fundamental Limits and Eigentasks," *Physical Review X*, vol. 13, no. 4, p. 041020, 2023.

[12] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Quantum Noise-Induced Reservoir Computing," 2022.

[13] A. Palacios de Luis, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Role of Coherence in Many-Body Quantum Reservoir Computing," *Communications Physics*, vol. 7, no. 1, p. 369, 2024.

[14] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Temporal Information Processing Induced by Quantum Noise," *Physical Review Research*, vol. 5, no. 2, p. 023057, 2023.

[15] R. Martínez-Peña and J.-P. Ortega, "Quantum Reservoir Computing in Finite Dimensions," *Physical Review E*, vol. 107, no. 3, p. 035306, 2023.

[16] P. Mujal, R. Martínez-Peña, G. L. Giorgi, M. C. Soriano, and R. Zambrini, "Time-Series Quantum Reservoir Computing with Weak and Projective Measurements," *npj Quantum Information*, vol. 9, p. 16, 2023.

[17] Y. Suzuki, Q. Gao, K. C. Pradel, K. Yasuoka, and N. Yamamoto, "Natural Quantum Reservoir Computing for Temporal Information Processing," *Scientific Reports*, vol. 12, p. 1353, 2022.

[18] A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Dissipation as a Resource for Quantum Reservoir Computing," *Quantum*, vol. 8, p. 1291, 2024.

[19] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, and T. A. Ohki, "Nonlinear Input Transformations Are Ubiquitous in Quantum Reservoir Computing," *Neuromorphic Computing and Engineering*, vol. 2, no. 1, p. 014008, 2022.

[20] R. Martínez-Peña, J. Nokkala, G. L. Giorgi, R. Zambrini, and M. C. Soriano, "Information Processing Capacity of Spin-Based Quantum Reservoir Computing Systems," *Cognitive Computation*, vol. 15, no. 5, pp. 1440–1451, 2023.

[21] W. D. Kalfus, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, T. A. Ohki, and L. C. G. Govia, "Hilbert Space as a Computational Resource in Reservoir Computing," *Physical Review Research*, vol. 4, no. 3, p. 033007, 2022.

[22] D. Fry, A. Deshmukh, S. Y.-C. Chen, V. Rastunkov, and V. Markov, "Optimizing Quantum Noise-Induced Reservoir Computing for Nonlinear and Chaotic Time Series Prediction," *Scientific Reports*, vol. 13, p. 19326, 2023.

[23] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, and T. A. Ohki, "Quantum Reservoir Computing with a Single Nonlinear Oscillator," *Physical Review Research*, vol. 3, no. 1, p. 013077, 2021.

[24] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data Re-Uploading for a Universal Quantum Classifier," *Quantum*, vol. 4, p. 226, 2020.

[25] M. C. Mackey and L. Glass, "Oscillation and Chaos in Physiological Control Systems," *Science*, vol. 197, no. 4300, pp. 287–289, 1977.

[26] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, vol. 57, no. 5, pp. 397–398, 1976.

[27] N. A. Gershenfeld and A. S. Weigend, "The Future of Time Series," Santa Fe Institute, Tech. Rep. 93-08-053, 1993.

[28] SILSO World Data Center, "Monthly mean total sunspot number V2.0," Royal Observatory of Belgium, Brussels, 2026, dataset SN_m_tot_V2.0, series 1749–2026, accessed July 2026, CC BY-NC 4.0.